

Assignment - 02

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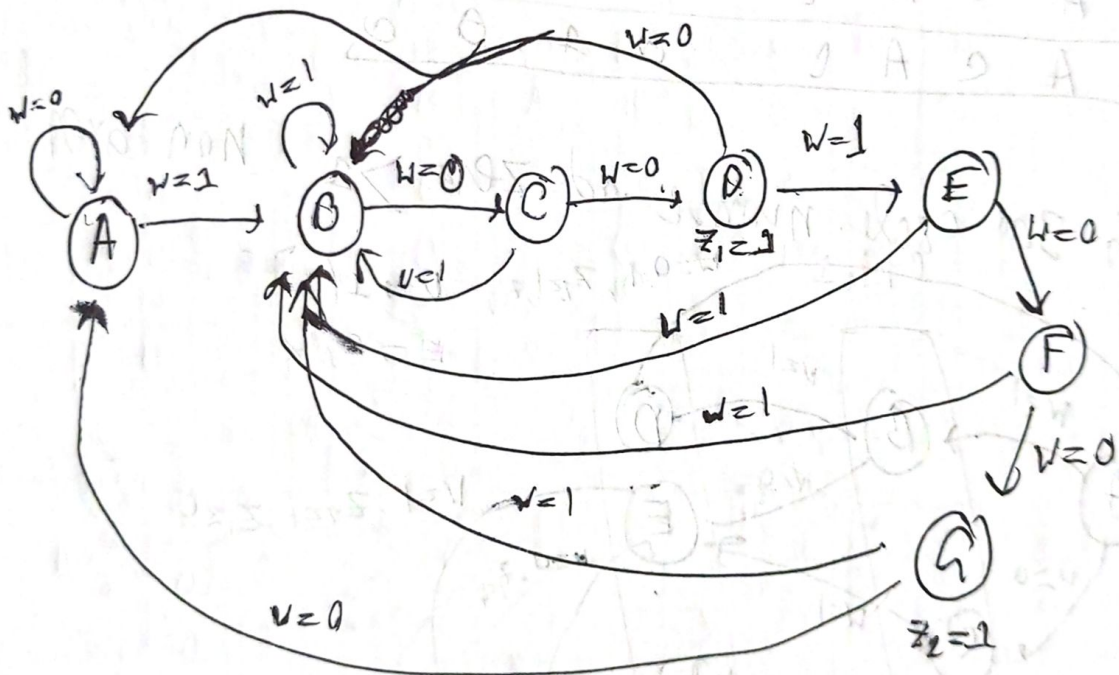
Section : 07

FRM 25

Set H

1. (a) '100'

~~odd~~ $odd \rightarrow Z_1$
 $even \rightarrow Z_2$



number of
 flip flops required, is 3.

b)

Present State $y_2 y_1 y_0$	Next state $y_2 y_1 y_0$		Output	
	$w=0$	$w=1$	z_1	z_2
A 000	A 000	D 001	0	0
B 001	C 010	B 001	0	0
C 010	D 011	B 001	0	0
D 011	A 000	E 100	0	1
E 100	F 101	B 001	0	0
F 101	G 110	B 001	0	1
G 110	A 000	B 001	1	0
ddd	dd	dd	dd	dd

c)

Output z_1, z_2

z_1	00	01	11	10
0	0	0	1	0
1	0	0	1	0

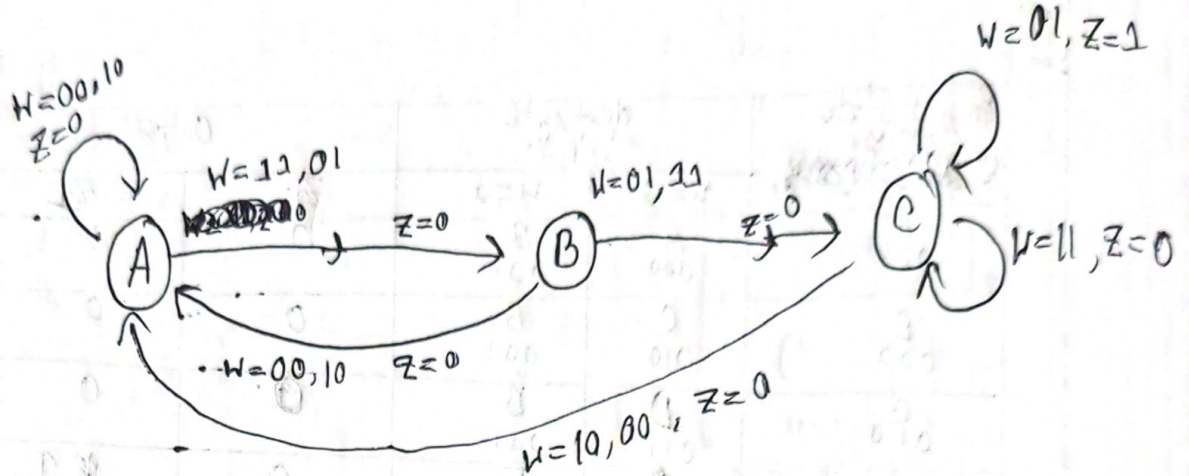
~~$z_1 = y_2 y_1$~~

$$z_1 = y_2 y_1$$

z_2	00	01	11	10
0	0	0	0	0
1	0	0	1	1

$$z_2 = y_2 y_1$$

2
(a)



Mealy FSM required

(b)

Present state $y_1 y_0$	Next state $Y_1 Y_0$				Output Z			
	00	01	10	11	00	01	10	11
A 00	A 00	B 01	A 00	B 01	0	0	0	0
B 01	A 00	C 10	A 00	C 10	0	0	0	0
C 10	A 00	C 10	A 00	C 10	0	1	0	0
11	dd	dd	dd	dd	dd	dd	dd	dd

(c)

Y_1

$y_1 y_0$	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	0	1	1	1
10	0	1	1	0

$$Y_1 = y_1 y_0 + \bar{y}_1 y_0$$

$$= y_0 (y_1 + \bar{y}_1)$$

$y_1 y_0$	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$Y_0 = \bar{y}_1 y_0$$

$y_1 y_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	1	0	0
10	0	1	0	0

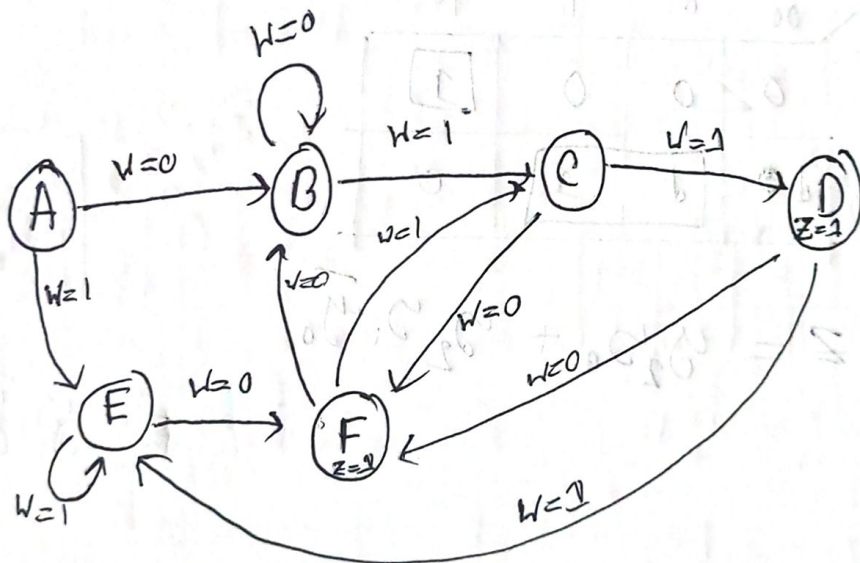
$$Z = x \bar{y}_1 y_0$$

Summer 2025 Set B

1

1a Moore FSM

1b



(c)

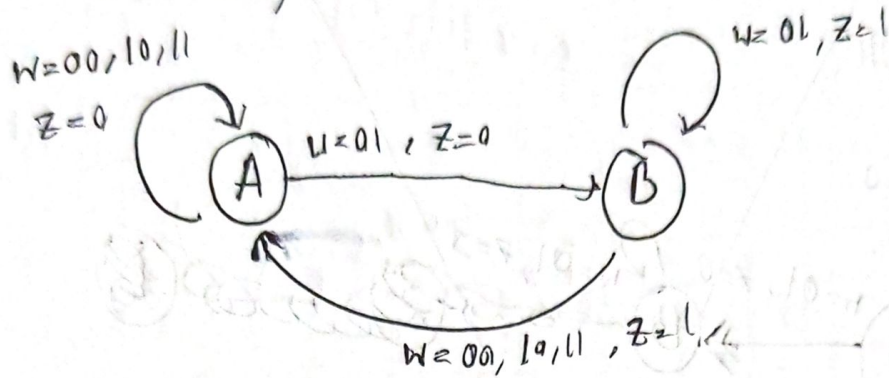
Present state $y_2 y_1 y_0$	Next state $y_2 y_1 y_0$		Output Z	
	$W=0$	$W=1$	00000000	
000 A	B 001	E 110	0	0
001 B	B 001	C 011	0	0
011 C	F 111	D 010	1	0
010 D	F 111	E 110	1	1
110 E	F 111	E 110	1	0
111 F	B 001	C 011	0	1
101	dd	dd	dd	dd
100	dd	dd	dd	dd

(d)

$y_2 \backslash y_1$	00	01	11	10
0	0	0	0	1
1	0	1	1	0

$$Z = y_2 y_1 + y_2 y_1 \bar{y}_0$$

2. (a) Mealy type FSM



(b)

Present state z_1	Next state (Y_1)				Output			
	00	01	10	11	00	01	10	11
A (0)	A 0	B 1	A 0	A 0	0	0	0	0
B (1)	A 0	B 1	A 0	A 0	0	1	0	0

(c)

Y_1

z_1	00	01	11	10
0	0	1	0	0
1	0	1	0	0

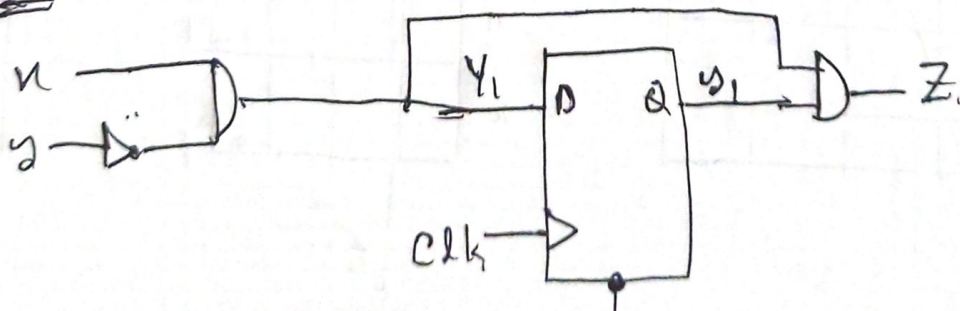
Z

z_1	00	01	11	10
0	0	0	0	0
1	0	1	0	0

$$Y_1 = \bar{y}_1 n$$

$$Z = y_1 \bar{y}_1 n$$

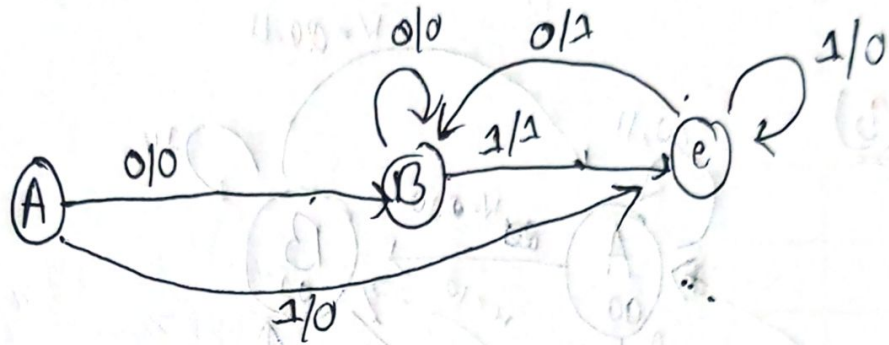
(d) Flip Flop required 1.



Spring 2025

Set A

2. (a)



(b)

Present state $y_1 y_0$	Next state ($y_1 y_0$)		Output	
	$w=0$	$w=1$	$w=0$	$w=1$
A 00	B 01	c 10	0	0
B 01	B 01	c 10	0	1
c 10	B 01	c 10	1	0
11	dd	dd	dd	dd

(c)

Y_1

w	$y_1 y_0$	00	01	11	10
0		0	0	1	0
1		1	1	1	1

$$Y_1 = w$$

Y_0

w	$y_1 y_0$	00	01	11	10
0		1	1	1	1
1		0	0	1	0

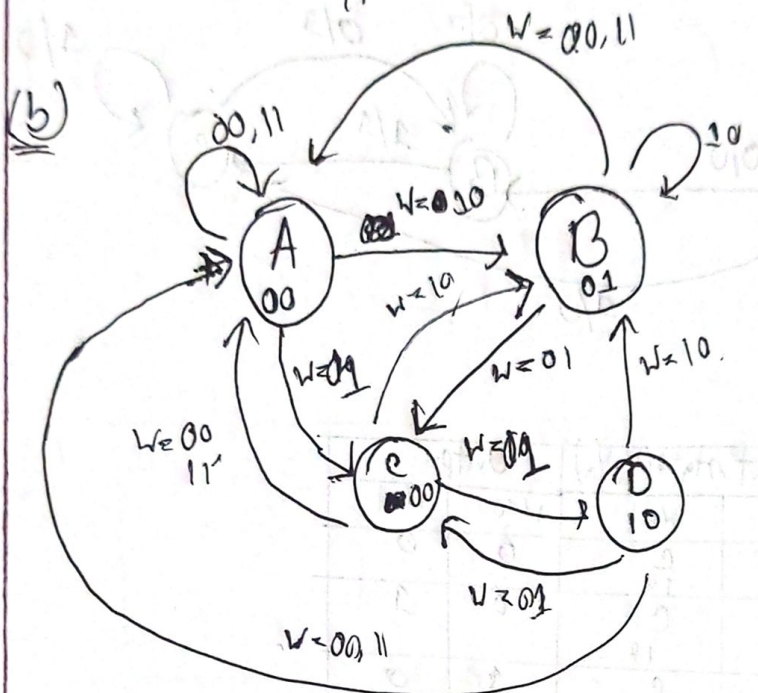
$$Y_0 = \bar{w}$$

Z

w	$y_1 y_0$	00	01	11	10
0		0	0	1	1
1		0	1	1	0

$$Z = w y_0 + \bar{w} y_1$$

1. (a) Moore type FSM



(c)

Present state $y_1 y_0$	Next state ($y_1 y_0$)				Output	
	00	01	10	11	Z_2	Z_1
A 00	A 00	C 11	B 01	A 00	0	0
B 01	A 00	C 11	B 01	A 00	0	1
C 11	A 00	D 10	B 01	A 00	0	0
D 10	A 00	C 11	B 01	A 00	1	0

(d) Number of flip-flops required = 2.

Z_2

$y_0 \backslash y_1$	0	1
0	0	0
1	1	1

$$Z_2 = y_0 \cdot \bar{y}_1$$

Z_1

$y_0 \backslash y_1$	0	1
0	0	1
1	0	0

$$Z_1 = \bar{y}_0 \cdot y_1$$